NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1513
Additional Practice Set 2

1. Consider the following matrices:

$$\boldsymbol{A}=\begin{pmatrix}0&1&1\\1&0&1\\1&1&0\end{pmatrix},\qquad \boldsymbol{B}=\begin{pmatrix}-1&3&-4\\2&4&1\\-4&2&-9\end{pmatrix},\qquad \boldsymbol{C}=\begin{pmatrix}2&0&1\\0&2&-1\\0&-1&2\end{pmatrix},$$

   (a) Determine whether each matrix above is non-singular. For those that are non-singular, find their inverses using Gauss-Jordan elimination.

   (b) Find the inverses of $\boldsymbol{A}^T, \boldsymbol{B}^T, \boldsymbol{C}^T$ if they exist.

   (c) Which of the homogeneous systems $\boldsymbol{Ax}=\boldsymbol{0}, \boldsymbol{Bx}=\boldsymbol{0}, \boldsymbol{Cx}=\boldsymbol{0}$ have non-trivial solutions? Find all the non-trivial solutions for these systems.

   (d) Without performing Gaussian Elimination, can you tell whether the homogeneous systems $\boldsymbol{ABx}=\boldsymbol{0}$ and $\boldsymbol{ACx}=\boldsymbol{0}$ have non-trivial solutions? Why?

2.

$$\boldsymbol{A}=\begin{pmatrix}1&2&0\\0&1&1\\-1&3&6\\2&1&0\\3&1&-1\end{pmatrix},\quad \boldsymbol{B}=\begin{pmatrix}2&1&4&1&2\\4&2&2&3&2\\2&1&-2&2&0\\6&3&6&4&4\end{pmatrix},\quad \boldsymbol{C}=\begin{pmatrix}1&4&5&8\\-1&4&3&0\\2&0&2&1\end{pmatrix}.$$

   (a) Determine the rank of each of the above matrices by finding their row echelon forms.

   (b) (**MATLAB**) Use the MATLAB command **rank** to check your answers in part (a).

   (c) Which of these matrices are full rank?

   (d) Is it possible to add a row to $\boldsymbol{C}$ so that the rank of the resulting 4×4 matrix is greater than $\operatorname{rank}(\boldsymbol{C})$? Why?

   (e) Explain why it s not possible to add another row to $\boldsymbol{A}$ such that the rank of the resulting matrix is greater than $\operatorname{rank}(\boldsymbol{A})$.

3. Let $\boldsymbol{A}=\begin{pmatrix}1513&1513&1513\\0&1513&1513\\0&0&1513\end{pmatrix}$ and $\boldsymbol{B}=\begin{pmatrix}1513&1513&1513\\1512&1512&1512\\1511&1511&1511\end{pmatrix}$.

   (i) Determine whether $\boldsymbol{A}$ and $\boldsymbol{B}$ are non-singular. Justify your answers.

   (ii) Find $\det\left(\frac{1}{1513}\boldsymbol{A}\right)$.

   (iii) Does $\boldsymbol{ABx}=\boldsymbol{0}$ have a non-trivial solution? Why?

4. A company manufactures three types of furniture (A, B, C) and the available resources in the company are: 100 units of labour, 60 units of upholstery and 200 units of wood. The various products require the following amounts of resources.

	A	B	C
Labour	1	1	2
Upholstery	0	1	2
Wood	3	2	2

   (i) Find the inverse of the production matrix for the company to produce the three types of furniture.

   (ii) Determine the number of units of each type of furniture that should be produced if the company wants to use up all its resources in its production schedule.

   (iii) What is the smallest and largest number of units of labour needed in order for the company to use up all its resources in its production schedule, if the available amount of the other two resources remain the same as above?

5. Consider the linear span in $\mathbb{R}^3$ given by $\text{span}\{(1,1,1),(3,0,1)\}$. This gives a plane P in the xyz-space.

   (i) Find a linear equation that represents the plane P.

   (ii) Does the vector $(24, 9, 14)$ lie on the plane P?

   (iii) Show that the vector $\boldsymbol{v} = (1, 2, -3)$ is orthogonal to both $(1, 1, 1)$ and $(3, 0, 1)$. Is $\boldsymbol{v}$ orthogonal to every vector in the plane?

6. Given the following sets of vectors in $\mathbb{R}^2, \mathbb{R}^3$ and $\mathbb{R}^4$ respectively:
   (a) $\boldsymbol{u} = (1, -1)$ and $\boldsymbol{v} = (-1, 3)$.
   (b) $\boldsymbol{u} = (1, 2, 3)$, $\boldsymbol{v} = (0, -3, 2)$ and $\boldsymbol{w} = (2, -1, 0)$.
   (c) $\boldsymbol{u} = (1, -1, 1, -1)$, $\boldsymbol{v} = (2, 1, 1, 2)$ and $\boldsymbol{w} = (3, 0, 2, 1)$.

   (i) Find the distance and angle between each pairs of vectors in (a), (b) and (c) separately.

   (ii) Which of the pairs are orthogonal to each other?

   (iii) For (b) and (c), determine whether $\boldsymbol{w}$ is a linear combination of $\boldsymbol{u}$ and $\boldsymbol{v}$.

   (iv) For (b) and (c), determine whether $\text{span}\{\boldsymbol{u}, \boldsymbol{v}\} = \text{span}\{\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}\}$.